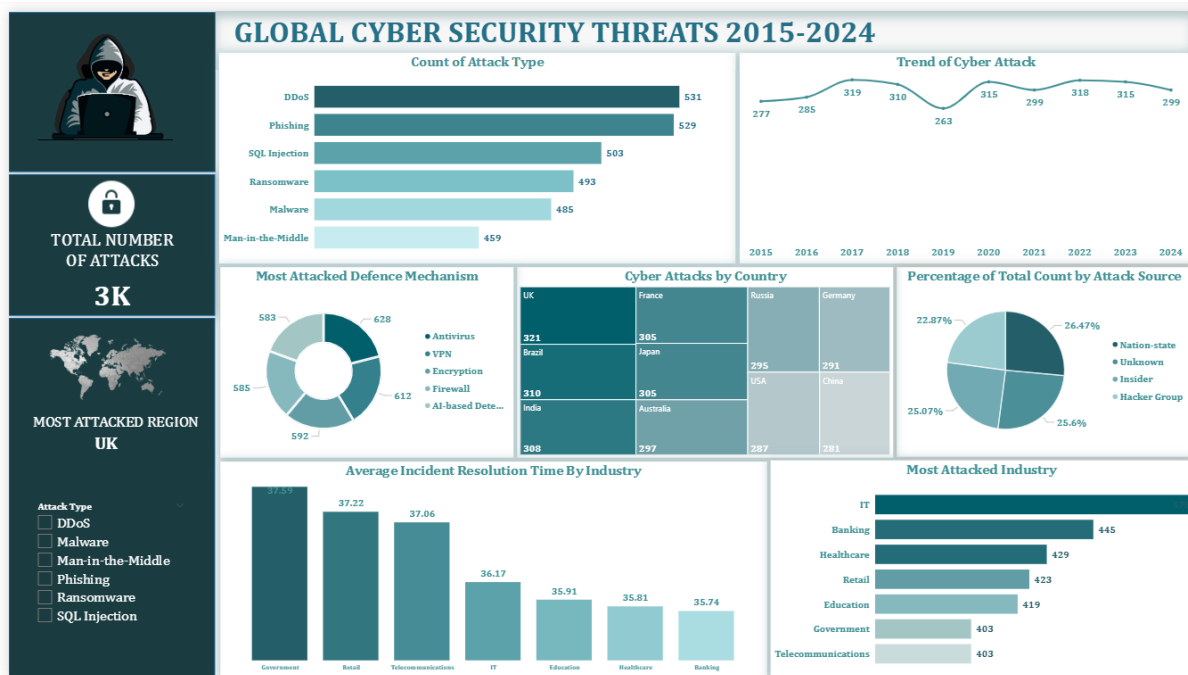




Dashboard of the Project

INTRODUCTION

This project is a deep look into how cyber-attacks have changed over the last 10 years (2015–2024). I wanted to find out three main things: who is behind the attacks, how they are doing it, and which countries are being hit the hardest.

PROJECT OVERVIEW

The Dataset

The heart of this project is a comprehensive dataset covering over 3,000 recorded cyber incidents between 2015 and 2024. It includes detailed information on attack types, targeted industries, and the defense mechanisms in place at the time of the breach.

What I Built

I developed an interactive Power BI dashboard that allows users to explore threat data through several key lenses:

- **Global Impact:** A geographic breakdown showing which countries, like the UK and Brazil, face the highest volume of attacks.
- **Attack Methods:** A ranking of the most frequent tactics, including DDoS and Phishing campaigns.

- **Industry Trends:** A comparison of how different sectors, such as IT and Banking, manage and resolve security incidents.
- **Time Analysis:** A 10-year trend line showing how the frequency of attacks has fluctuated over the last decade.

PROBLEM STATEMENT

The Challenge

In the modern digital landscape, cybersecurity data is often massive and unorganized. For this project, the raw data consisted of over 3,000 individual attack records, making it nearly impossible to identify critical patterns or trends manually.

Without a clear way to visualize this information, several major problems existed:

- **Invisible Trends:** It was unclear if global cyber-attacks were increasing or decreasing over the 10-year period (2015–2024).
- **Target Blindness:** There was no easy way to see which specific countries or industries were facing the most threats, such as the high volume of attacks in the UK.
- **Defense Gaps:** It was difficult to measure how effectively different industries were resolving incidents, leaving some sectors more vulnerable than others.

The Solution

I developed this Power BI dashboard to transform this messy data into a single source of truth.

- **Data Transformation:** I used Power Query to clean and structure the data, ensuring all attack types and locations were standardized.
- **Interactive Analysis:** I implemented dynamic slicers and DAX measures (like "Final Count") to ensure the dashboard remains accurate and responsive, even when filtering for specific threats like DDoS or Phishing.
- **Strategic Visualization:** By using modern visuals like Treemaps and Trend Lines, I created a layout that allows anyone to understand the global threat level in seconds.

TOOL USED

Data Preparation & Cleaning

- **Microsoft Excel:** Used to perform the initial data cleaning, ensuring all 3,000+ records were formatted correctly before analysis.
- **Data Standardizing:** Organized columns for Attack Types, Countries, and Industry sectors to ensure a smooth import process.

Visualization & Dashboarding

- **Microsoft Power BI:** Used to build the interactive dashboard and map the global threat landscape.
- **DAX (Data Analysis Expressions):** Developed custom measures to keep the charts accurate when using the "Attack Type" slicer.
- **Interactive Design:** Configured cross-filtering and visual interactions so that every chart responds to user selections in real-time.

PRE ANALYSIS

Before I started making the charts, I spent time planning how to turn the data into a clear story. This phase helped me understand the data better and decide which parts were the most important to show.

- Organizing the Data

Finding Relationships: I looked at how different parts of the data, like "Attack Type" and "Country," connected to each other.

Asking the Right Questions: I identified the main questions I wanted to answer, such as which industries are hit the hardest and if attacks are increasing over time.

- Shaping the Story

Better Storytelling: I planned the dashboard to flow from a big-picture view (Total Attacks) down to specific details (Attack Types and Locations).

Target Audience: I made sure the dashboard would be easy to understand for both technical experts and business managers who just want a quick summary.

- Strategic Planning

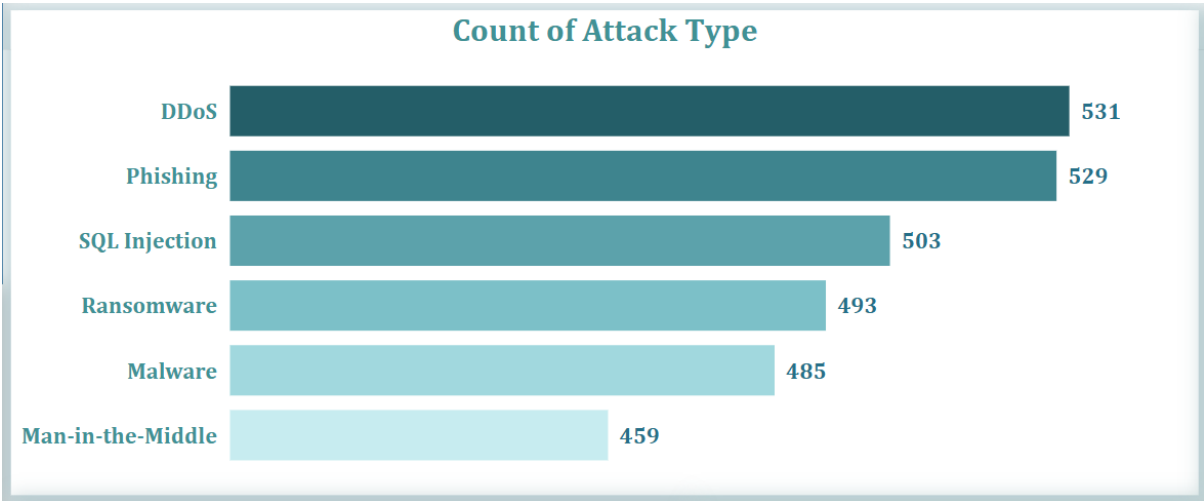
Choosing Visuals: Based on my early thoughts, I picked specific charts, like the Treemap and Line Chart, to make the data easy to read at a glance.

Focusing on Value: This step helped me focus on the most valuable insights so the dashboard wouldn't feel cluttered or confusing.

ANALYSIS OF THE CHARTS

The analysis revealed some compelling insights into the state of global cybersecurity, broken down by specific data visualizations:

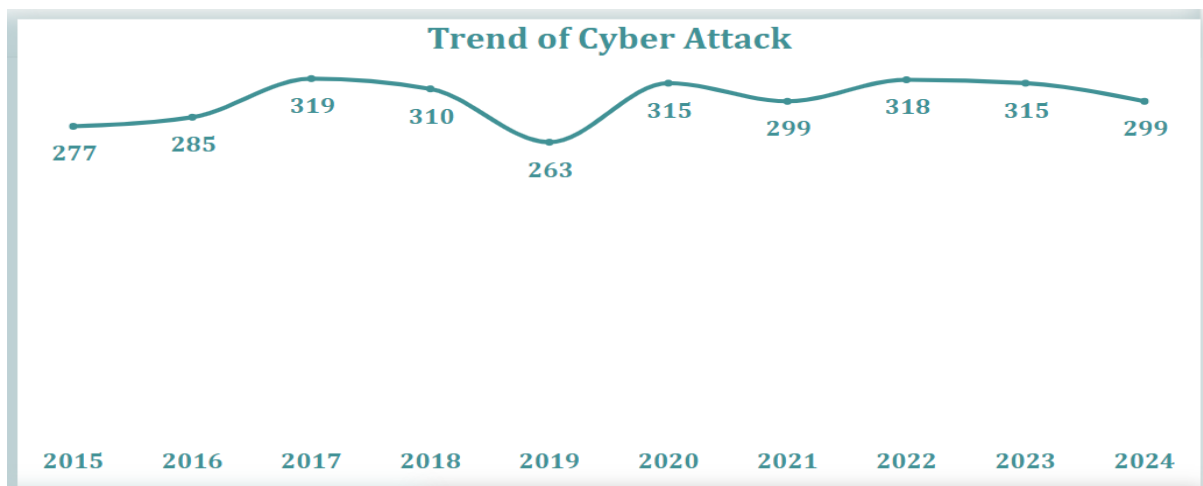
COUNT OF ATTACK TYPE



This bar chart shows which methods hackers used most often between 2015 and 2024. The data reveals that DDoS (531 attacks) and Phishing (529 attacks) are the top threats, followed closely by SQL Injection (503 attacks). These high numbers show that hackers still rely heavily on overwhelming networks and tricking users to gain access.

On the lower end of the scale, Ransomware (493) and Malware (485) remain constant threats, while Man-in-the-Middle attacks were the least frequent at 459 incidents. Overall, the chart proves that while attack methods vary, the most basic and high-volume techniques like DDoS and Phishing continue to be the biggest challenges for digital security.

TREND OF CYBER ATTACK



This line chart visualizes the trend of cyber attacks from 2015 to 2024.

It shows a steady rise in cyber attacks from 2015 to 2017, indicating increasing digital threats.

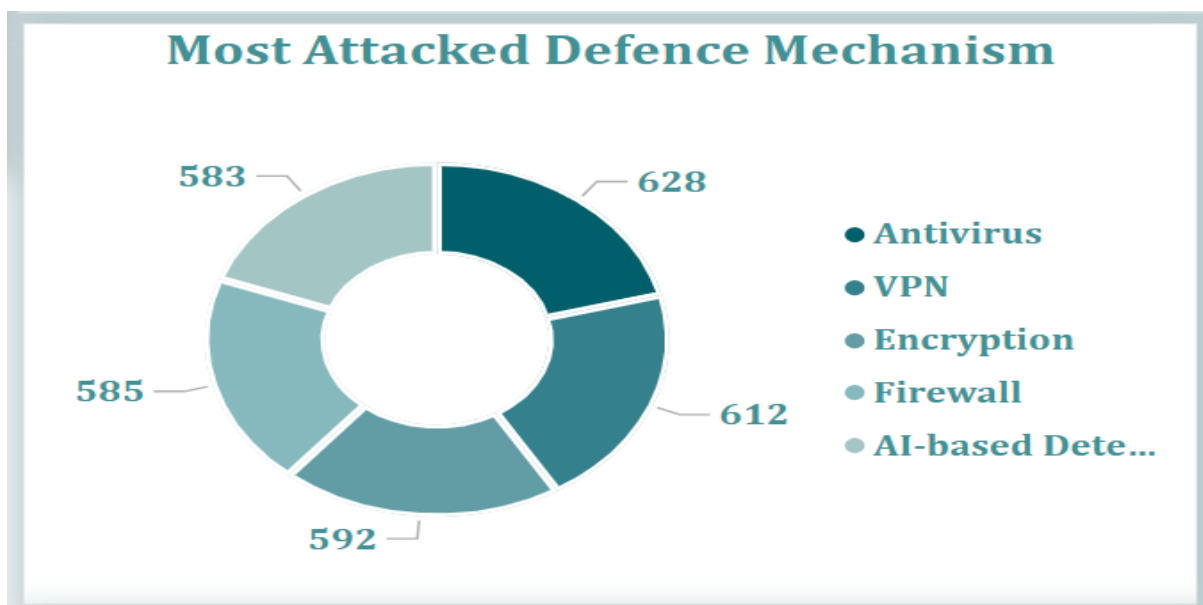
In 2019, the number of attacks drops significantly, showing a temporary decline.

After 2019, cyber attacks increase again, with noticeable peaks in 2020 and 2022.

Although there are minor fluctuations in later years, the overall trend remains high.

This analysis highlights the growing importance of cybersecurity measures over time.

MOST ATTACKED DEFENCE MECHANISM



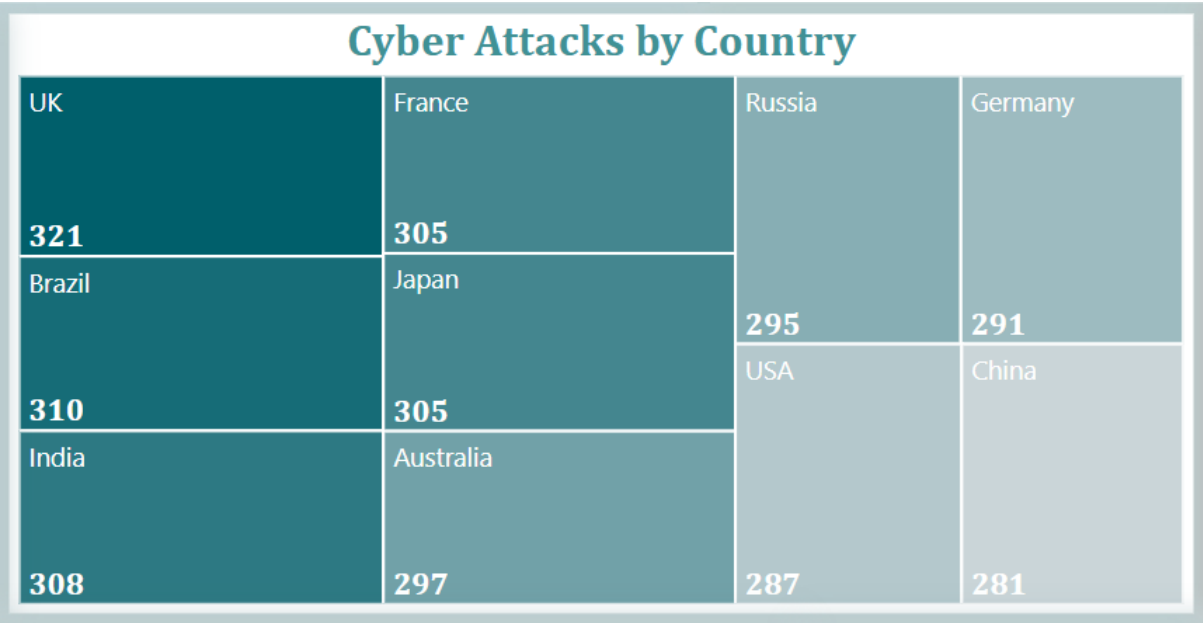
This doughnut chart represents the most attacked defence mechanisms in cybersecurity systems.

It shows that Antivirus systems face the highest number of attacks, followed closely by VPNs. Encryption and Firewall mechanisms also experience a significant number of attacks. AI-based detection systems have comparatively fewer attacks but are still targeted by cyber threats.

The distribution highlights that commonly used security tools are more frequently attacked.

Overall, the chart emphasizes the need to strengthen all defence mechanisms, especially widely deployed ones.

CYBER ATTACKS BY COUNTRY



This tree map visualizes the distribution of cyber attacks across different countries.

The size of each block represents the number of cyber attacks reported by that country.

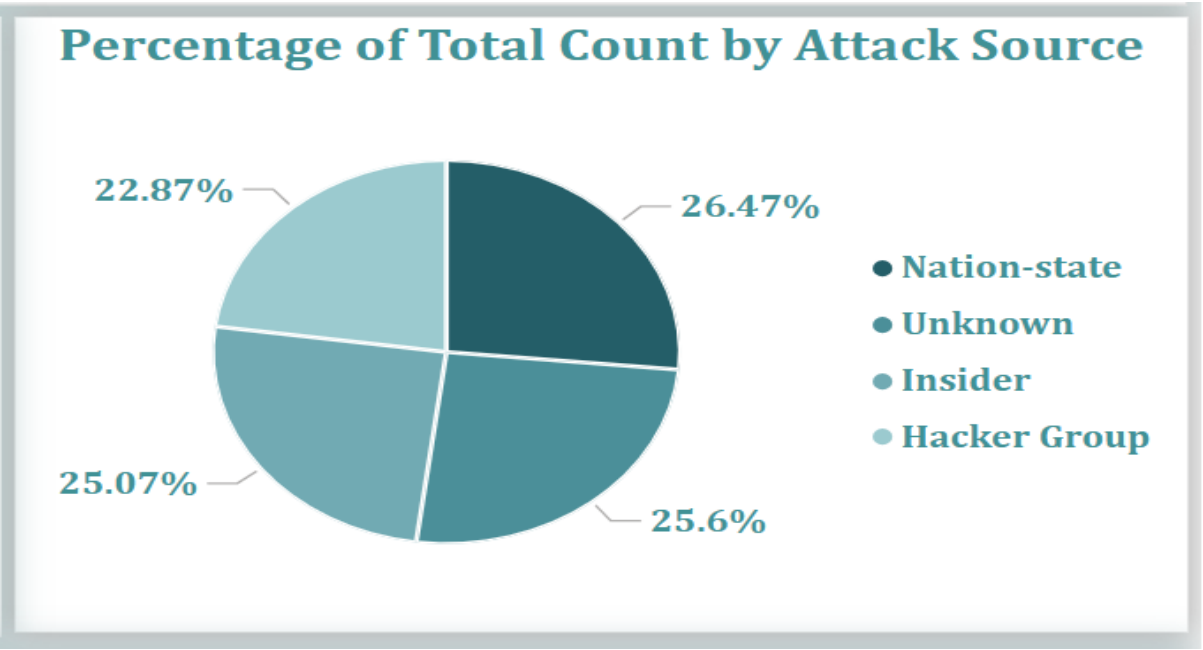
The UK shows the highest number of cyber attacks, followed by Brazil, India, and France.

Countries like Japan and Australia also report a significant number of attacks.

The USA, Russia, Germany, and China show comparatively lower but notable attack counts.

Overall, the chart highlights how cyber threats are globally distributed across multiple countries.

PERCENTAGE OF TOTAL COUNT BY ATTACK SOURCE



This pie chart represents the percentage distribution of cyber attacks by attack source.

Nation-state attacks account for the largest share of total attacks, indicating organized and targeted threats.

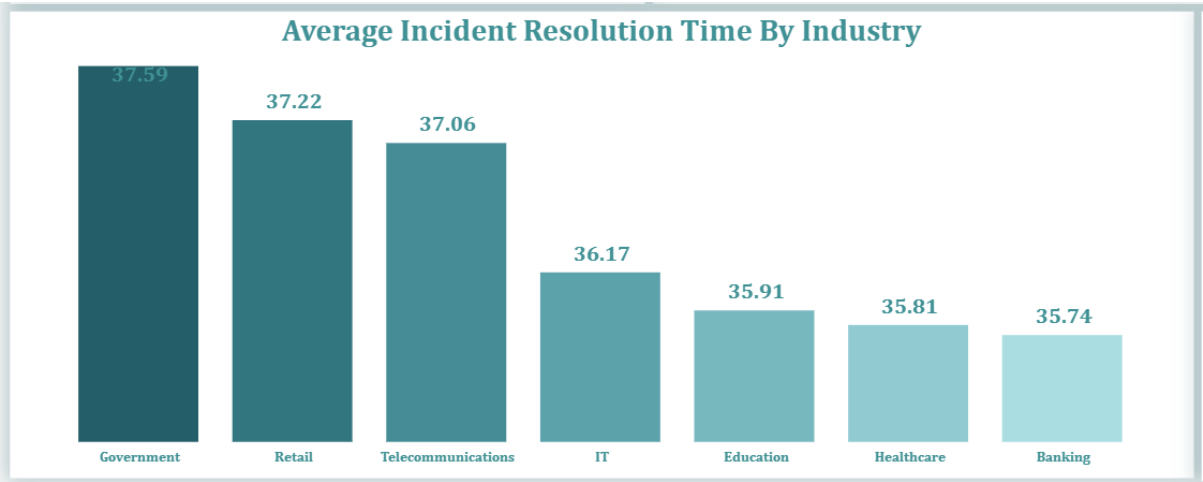
Unknown sources also contribute a significant portion, highlighting challenges in attack attribution.

Insider threats form a considerable percentage, showing risks from within organizations.

Hacker groups make up the remaining share but still pose a serious security concern.

Overall, the chart shows that cyber attacks originate from diverse sources, requiring multi-layered security strategies.

AVERAGE INCIDENT RESOLUTION TIME BY INDUSTRY



This bar chart shows the average time taken to resolve cybersecurity incidents across different industries.

The government sector has the highest resolution time, indicating more complex or sensitive incidents.

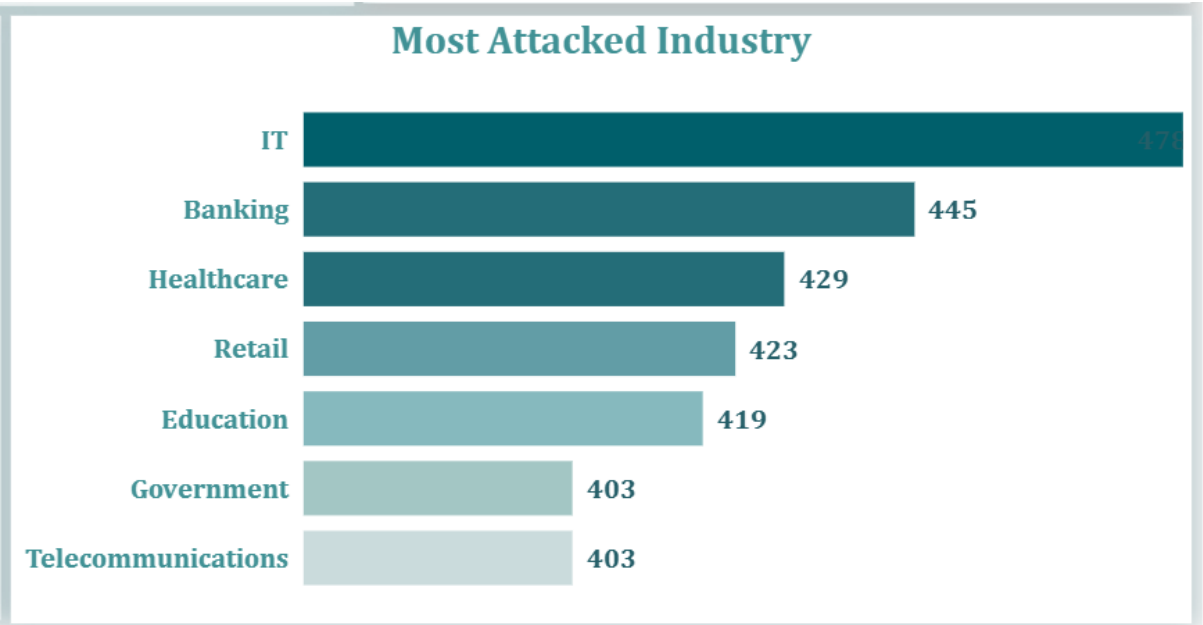
Retail and telecommunications industries also show relatively higher resolution times.

IT, education, and healthcare sectors resolve incidents faster compared to others.

The banking industry has the lowest average resolution time, showing more efficient incident handling.

Overall, the chart highlights variations in cybersecurity response efficiency across industries.

MOST ATTACKED INDUSTRY



This horizontal bar chart shows the number of cyber attacks across different industries.

The IT industry is the most attacked sector, indicating its high exposure to cyber threats.

The banking and healthcare industries also experience a large number of attacks.

Retail and education sectors show moderate attack levels.

Government and telecommunications industries report comparatively fewer attacks.

Overall, the chart highlights that technology-driven industries are primary targets for cyber attackers.

KEY INSIGHTS

- The dashboard shows a total of 3,000 cyber attack incidents recorded between 2015 and 2024, highlighting the scale of global cybersecurity threats.
- DDoS, Phishing, and SQL Injection are the most frequent attack types, indicating that attackers continue to exploit widely known and effective attack vectors.
- The trend of cyber attacks fluctuates over the years rather than showing a steady rise, reflecting a constantly evolving cybersecurity landscape where attackers and defenders adapt to each other.
- The UK emerges as the most attacked region, followed by countries such as Brazil, India, Japan, and France, suggesting a higher concentration of digital assets or vulnerabilities in these regions.
- Antivirus, Firewalls, and AI-based detection systems are the most targeted defense mechanisms, indicating that attackers actively attempt to bypass core security layers.
- Analysis of attack sources shows a nearly balanced contribution from Nation-state, Insider, Hacker Groups, and Unknown sources, highlighting the complexity of threat attribution.
- The IT, Banking, and Healthcare industries are the most attacked sectors, primarily due to the sensitive data they manage and the critical services they provide.
- Incident resolution time varies across industries, with Government and Retail sectors taking longer on average, indicating differences in incident response maturity and operational readiness.

CONCLUSIONS

This project provides a comprehensive analysis of global cybersecurity threats from 2015 to 2024 using interactive data visualizations. The findings show that cyber attacks remain persistent and continue to target widely used systems, industries, and regions. Attack types such as DDoS, Phishing, and SQL Injection dominate the threat landscape, indicating that traditional attack vectors are still highly effective.

The fluctuating trend of attacks over the years reflects the dynamic nature of cybersecurity, where improvements in defense mechanisms are met with evolving attack strategies. The concentration of attacks in specific countries and industries highlights uneven global risk exposure and the need for region- and sector-specific security measures.

Additionally, the frequent targeting of core defense mechanisms and the variation in incident resolution times across industries emphasize gaps in security readiness and response

capabilities. Overall, this analysis underscores the importance of continuous monitoring, advanced threat detection, and stronger cybersecurity strategies to mitigate future risks.